

Solutions to Week 3 Tutorial Questions (Ch2)

Chapter 2:

Q1: Macroeconomics P.51 Review Questions #1

Q2: Macroeconomics P.51 Review Questions #5

Q3: Macroeconomics P.52 Numerical Problems #5

Q4: Macroeconomics P.52 Numerical Problems #6

Q5: Macroeconomics P.53 Numerical Problems #8

Q6: Macroeconomics P.53 Analytical Problems #4

Q1: What are the three approaches to measuring economic activity? Why do they give the same answer?

Answer: *The three approaches to national income accounting are the product approach, the income approach, and the expenditure approach. They all give the same answer because they are designed that way; any entry based on one approach has an entry in the other approaches with the same value. Whenever output is produced and sold, its production is counted in the product approach, its sale is counted in the expenditure approach, and the funds received by the seller are counted in the income approach.*

Q2: Define private saving. How is private saving used in the economy? What is the relationship between private saving and national saving?

Answer: *Private saving is private disposable income minus consumption. Private disposable income is total output minus taxes paid plus transfers and interest received from the government. Private saving is used to finance investment spending, the government budget deficit, and the current account. National saving is private saving plus government saving.*

Q3: You are given the following information about an economy:

Gross private domestic investment	40
Government purchases of goods and services	30
Gross national product (GNP)	200
Current account balance	-20
Taxes	60
Government transfer payments	25

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Interest payment from the government (all to domestic households)	15
Factor income from the rest of the world	7
Factor payment to the rest of the world	9

Find the following, assuming that government investment is zero:

- a. Consumption
- b. Net exports
- c. GDP
- d. Net factor payments
- e. Private saving
- f. Government saving
- g. National saving

Answer: *Given data: $I = 40$, $G = 30$, $GNP = 200$, $CA = -20 = NX + NFP$, $T = 60$, $TR = 25$, $INT = 15$, $NFP = 7 - 9 = -2$.*

a. Consumption = 150

Since $GDP = GNP - NFP$, thus $GDP = Y = 200 - (-2) = 202$

Since $NX + NFP = CA$, thus $NX = CA - NFP = -20 - (-2) = -18$.

Since $Y = C + I + G + NX$, thus $C = Y - (I + G + NX) = 202 - (40 + 30 + (-18)) = 150$.

b. Net exports = -18

Since $CA = -20 = NX + NFP$ and $NFP = 7 - 9 = -2$.

Thus $NX = CA - NFP = -20 - (-2) = -18$

c. GDP = 202

Since $GDP = GNP - NFP$, thus $GDP = Y = 200 - (-2) = 202$

d. Net factor payments from abroad = -2

$NFP = 7 - 9 = -2$.

e. Private saving = 30

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$$Spvt = (Y + NFP - T + TR + INT) - C = (202 + (-2) - 60 + 25 + 15) - 150 = 30.$$

f. Government saving = -10

$$Sgovt = (T - TR - INT) - G = (60 - 25 - 15) - 30 = -10$$

g. National saving = 20

$$S = Spvt + Sgovt = 30 + (-10) = 20.$$

Q4: Consider an economy that produces only three types of fruit: apples, oranges, and bananas. In the base year (a few years ago), the production and price data were as follows:

Fruit	Quantity	Price
Apples	3000 bags	\$2 per bag
Bananas	6000 bunches	\$3 per bunch
Oranges	8000 bags	\$4 per bag

In the current year, the production and price data are as follows:

Fruit	Quantity	Price
Apples	4000 bags	\$3 per bag
Bananas	14000 bunches	\$2 per bunch
Oranges	32000 bags	\$5 per bag

<i>Base-year quantities at current-year prices:</i>		<i>at base-year prices:</i>
<i>Apples</i>	<i>3 000 x \$3 = \$ 9 000</i>	<i>3000 x \$2 = \$ 6 000</i>
<i>Bananas</i>	<i>6 000 x \$2 = \$12 000</i>	<i>6000 x \$3 = \$18 000</i>
<i>Oranges</i>	<i>8 000 x \$5 = \$40 000</i>	<i>8000 x \$4 = \$32 000</i>
<i>Total</i>	<i>\$61 000</i>	<i>\$56 000</i>
<i>Current-year quantities at current-year prices:</i>		<i>at base-year prices:</i>
<i>Apples</i>	<i>4 000 x \$3 = \$ 12 000</i>	<i>4 000 x \$2 = \$ 8 000</i>
<i>Bananas</i>	<i>14 000 x \$2 = \$ 28 000</i>	<i>14 000 x \$3 = \$ 42 000</i>
<i>Oranges</i>	<i>32 000 x \$5 = \$160 000</i>	<i>32 000 x \$4 = \$128 000</i>
<i>Total</i>	<i>\$200 000</i>	<i>\$178 000</i>

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- a. Find nominal GDP in the current year and in the base year. What is the percentage increase since the base year?

Answer: *Nominal GDP is just the dollar value of production in a year at prices in that year. Nominal GDP is \$56 000 in the base year and \$200 000 in the current year. Nominal GDP grew 257% between the base year and the current year: $[(\$200\,000 / \$56\,000) - 1] \times 100\% = 257\%$.*

- b. Find real GDP in the base year and in the current year. By what percentage does real GDP increase from the base year to the current year?

Answer: *Real GDP is calculated by finding the value of production in each year at base-year prices. Thus, from the table above, real GDP is \$56 000 in the base year and \$178,000 in the current year. In percentage terms, real GDP increases from the base year to the current year by $[(\$178\,000 / \$56\,000) - 1] \times 100\% = 218\%$.*

- c. Find the GDP deflator for the current year and the base year. By what percentage does the price level change from the base year to the current year?

Answer: *The GDP deflator is the ratio of nominal GDP to real GDP. In the base year, nominal GDP equals real GDP, so the GDP deflator is 1. In the current year, the GDP deflator is $\$200\,000 / \$178\,000 = 1.124$. Thus the GDP deflator changes by $[(1.124 / 1) - 1] \times 100\% = 12.4\%$ from the base year to the current year.*

- d. Would you say that the percentage increase in nominal GDP in this economy since the base year is due more to increases in prices or increases in the physical volume of output?

Answer: *Nominal GDP rose 257%, prices rose 12.4%, and real GDP rose 218%, so most of the increase in nominal GDP is because of the increase in real output, not prices.*

Q5: Hy Marks buys a one-year government bond on January 1, 2010, for \$500. He receives principal plus interest totaling \$545 on January 1, 2011. Suppose that the CPI is 200 on January 1, 2011, and 214 on January 1, 2012. This increase in prices is more than Hy had anticipated; his guess was that the CPI would be at 210 by the beginning of 2011.

Find the nominal interest rate, the inflation rate, the real interest rate, Hy's expected inflation rate, and Hy's expected real interest rate.

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Answer: *The nominal interest rate is $[(545 / 500) - 1] \times 100\% = 9\%$. The inflation rate is $[(214 / 200) - 1] \times 100\% = 7\%$. So the real interest rate is 2% (9% nominal rate – 7% inflation rates). Expected inflation was only $[(214 / 210) - 1] \times 100\% = 1.9\%$, so the expected real interest rate was 7.1% (9% nominal rate – 1.9% expected inflation rate).*

Q6: Economists have tried to measure the GDPs of virtually all the world's nations. This problem asks you to think about some practical issues that arise in that effort.

- a. Before the collapse of the Soviet system the economies of the Soviet Union and Eastern Europe were centrally planned. One aspect of central planning is that most prices are set by the government. A government-set price may be too low in that people want to buy more of the good at the fixed price than there are supplies available; or the price may be too high so that large stocks of the good sit unsold on store shelves. During the past several years, central planning has been largely eliminated in Eastern Europe and the former Soviet Union, but government price-setting has not been completely abandoned. For example, Russia still keeps energy prices well below market-clearing levels.

What problem does government control of prices create for economists attempting to measure a country's GDP? Suggest a strategy for dealing with this problem.

Answer: *The problem in a planned economy is that prices do not measure market value. When the price of an item is too low, then goods are really more expensive than their listed price suggests—we should include in their market value the value of time spent by consumers waiting to make purchases. Because the item's value exceeds its cost, measured GDP is too low. When the price of an item is too high, goods stocked on the shelves may be valued too highly, This results in an overvaluation of firms' inventories, so that measured GDP is too high. A possible strategy for dealing with this problem is to have GDP analysts estimate what the market price should be (perhaps by looking at prices of the same goods in market economies) and use this "shadow" price in the GDP calculations.*

- b. In very poor agricultural countries, many people grow their own food, make their own clothes, and provide services for each other within a

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family or village group. Official GDP estimates for these countries are often extremely low, perhaps just a few hundred dollars per person. Some economists have argued that the official GDP figures underestimate these nations' actual GDPs. Why might this be so? Again, can you suggest a strategy for dealing with this measurement problem?

Answer: *"Homework" is not calculated in the GDP accounts because it is not sold on the market, making it difficult to measure. One way to do it might be to look at the standard of living relative to a market economy, and estimate what income it would take in a market economy to support that standard of living.*